LoomPilot: Application Workflow Document

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1. Purpose

This document describes how work moves through LoomPilot, from a client's order request to delivery: who does what, what each agent does, which statuses an order passes through, and which rules apply. It is written for the owner, the admin and supervisors, and for the development team.

2. Actors and responsibilities

Actor	Responsibilities
Client	Submits orders, accepts or declines quotations, asks questions in chat, tracks progress
Admin	Reviews feasibility and quotations, approves them, decides on purchases, handles discount requests and chat takeover, hands accepted orders to supervisors, tracks all orders
Supervisor	Submits the daily report (attendance, machines, material stock, production), receives assigned orders and runs production
Feasibility agent	Decides whether the deadline can be met, and explains why or why not
Quotation agent	Prepares the quotation and its explanation
Market price agent	Checks raw material prices every day and flags surges
Client assistant	Answers client questions in chat and escalates discount requests
Monitoring agent	Tracks order progress and deadline risk from supervisor reports

Agents advise and draft. The admin approves and decides.

3. End-to-end workflow at a glance

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Client submits order
   -> Admin notified, Feasibility agent runs
        |-- Feasible ------------> Quotation agent drafts quote
        |                              -> Admin reviews and approves
        |                              -> Quote sent to client
        '-- At risk or not feasible -> Admin sees reasons and recommendations
                                       (local market prices, overtime, extra
workers, later date, partial shipment)
                                       -> Admin decides: buy materials,
propose alternative, or decline
                                       -> Feasibility re-runs, then the quote
path above

Client responds:  Accept | Decline | Ask questions in chat (AI assistant,
discounts go to the admin)
   -> Accept: Admin notified -> Admin assigns the order to supervisors
   -> Supervisors produce and submit daily reports -> Monitoring agent
updates progress
   -> Admin and client see the phase and produced quantity
   -> Delivery and payment

Every day: Market price agent checks raw material prices
   -> Surge: affected quotes get a proposed revision -> Admin approves ->
Client notified
   -> Client asks why: AI assistant explains. Wants a discount: sent to the
admin.
```

4. Status lifecycles

4.1 Order

Status	Meaning	Moves to
submitted	Client sent the request	analyzing
analyzing	Feasibility agent is running	quotation_drafting (feasible), awaiting_admin_decision (at risk or not feasible), analysis_failed

Status	Meaning	Moves to
analysis_failed	Agent error or timeout	analyzing (retry)
awaiting_admin_decision	Admin must choose how to proceed	analyzing (after a purchase decision), quotation_drafting (alternative chosen), rejected_by_admin
quotation_drafting	Quotation agent is preparing the quote	quotation_pending_approval
quotation_pending_approval	Admin must review	quotation_sent, quotation_drafting (edit and recalculate), rejected_by_admin
quotation_sent	Client has the quote	client_accepted, declined_by_client, expired
client_accepted	Client accepted, admin notified	assigned, cancelled
assigned	Admin handed the order to supervisors	in_production
in_production	Supervisors are producing	ready_for_delivery, on_hold, cancelled
ready_for_delivery	Production complete	partially_delivered, delivered
partially_delivered	Some quantity shipped	delivered
delivered	All quantity shipped	completed
completed	Closed after delivery and payment	none

Status	Meaning	Moves to
declined_by_client, rejected_by_admin, expired, cancelled	Terminal states	none

4.2 Quotation, revision, feasibility and chat

Item	Statuses
Feasibility report	running, ready, failed, superseded
Quotation	generating, pending_approval, approved, sent, accepted, declined, rejected, expired, superseded
Price revision	proposed, approved, sent, accepted, declined, rejected_by_admin, superseded
Conversation	agent_handling, escalated, admin_handling, resolved
Escalation	open, in_progress, resolved
Alert	open, acknowledged, resolved, dismissed

Each status table is enforced in code. An illegal move returns an error and changes nothing.

5. Detailed workflows

5.1 Client submits an order

Client steps: log in, choose New order, fill in the form, submit.

Form fields: product (from the catalog, or described in text), quantity, required delivery date, specifications (sizes, colours, fabric, print or embroidery, packaging), delivery address, notes and optional reference files.

System steps:

1. Validate: quantity above zero, date in the future, required fields present.

2. Create the order with a unique number, status submitted. An idempotency key stops duplicates from double clicks.
3. Notify the admin (in-app and email), and queue the feasibility run. Status becomes analyzing.
4. The client sees 'Received, being analysed' with the order number.

5.2 Feasibility analysis

The feasibility agent gathers: the order, product routing and bill of materials, stock and reserved stock, machines and their availability, workers per process with 30-day average attendance, average output per worker, the remaining work of orders already confirmed, supplier lead times, holidays and working days, and settings.

Steps (all calculated in code):

1. Materials: required quantity including wastage versus available stock. A shortfall sets the material ready date to today plus the shortest supplier lead time.
2. Capacity per process: labour capacity is expected present workers times average output per worker. Machine capacity is available machines times hours times units per hour. Daily capacity is the lower of the two.
3. Free capacity: daily capacity minus the load of confirmed orders. Without this step every new order would look feasible.
4. Process days: quantity divided by free capacity, counting working days only, with configurable overlap between stages.
5. ETA: material ready date plus process days plus safety days.
6. Verdict: feasible when the ETA is before the deadline with the safety buffer intact, at risk when the ETA falls inside the buffer or major risk factors exist, not feasible when the ETA is after the deadline.
7. Risk factors recorded with a severity: falling attendance, machine downtime, material shortage, long supplier lead time, high price volatility, heavy committed load, holidays inside the window, and thin data.

Output: a feasibility report with the verdict, ETA, bottleneck process, risk factors, recovery options (when needed), a readable reasons summary written by the LLM, and a snapshot of the data used.

5.3 Feasible path: quotation

1. The quotation agent calculates material cost with wastage, processing cost, overhead, price-risk buffer from recent volatility, margin, final price, validity date and the ETA basis. The LLM writes the explanation and lists assumptions.
2. The quotation appears in the admin's queue as pending approval, next to the feasibility report.
3. The admin can edit margin, line items or the delivery date within the feasible range. The agent recalculates and the admin sees what changed.
4. The admin approves or rejects with a reason. The system warns when the margin is below the minimum set in Settings.
5. On approval the client version of the PDF is generated (no cost, wastage, risk or margin shown) and the client is notified. Status becomes quotation_sent.

The client never sees a quotation that the admin has not approved.

5.4 At risk or not feasible

The admin sees the reasons and recommendations. Recommendations include overtime hours needed, extra workers needed, the earliest feasible date, a partial shipment split, and buying materials from local suppliers or the local market with prices, lead times and the effect on ETA and price.

The admin chooses one of:

- Buy raw materials: pick a supplier, quantity and price. A purchase decision and supplier order are created. The feasibility agent re-runs with the expected arrival date, so the deadline and price update.
- Propose an alternative: a later date, partial shipment or reduced quantity. The quotation agent drafts a quote with those terms and the normal approval path follows.
- Decline the order: the admin gives a reason. The client is told in client-safe wording.

A purchase is a business decision. The system recommends and records it but never buys anything by itself.

5.5 Client response

Client action	Result
Accept	Acceptance is recorded (who, when, quote version). Order status client_accepted. Admin is notified.
Decline	Status declined_by_client, with an optional reason.
Ask a question	Opens the chat. The AI assistant answers (see 5.10).
No response	The quotation expires after its validity period. Expired quotes cannot be accepted.

5.6 Handover to supervisors

When the client accepts, the admin is notified and does the handover:

1. Review required materials against stock. If short, record a purchase decision.
2. Reserve stock for the order.
3. Assign one or more supervisors. Each is responsible for specified processes or stages.
4. Confirm the production plan taken from the quotation (stages, planned dates, daily targets).
5. Save. Status becomes assigned and the supervisors are notified.
6. A supervisor acknowledges and starts production. Status becomes in_production.

The handover task shows product, quantity, specifications, due date, process plan and materials allocated.

5.7 Supervisor daily report

The supervisor submits one report for each day, from a mobile-friendly form that also works offline.

Section	Content
Attendance	Each worker: present, absent or half day, with process and hours

Section	Content
Machines	Each machine: available, under maintenance or breakdown, hours available
Materials	Stock counts or consumption of key materials
Production	For each assigned order and process: quantity produced and defective
Issues	Optional: breakdown, shortage, delay, with notes

Rules: one report per supervisor per day. Corrections need a reason and are logged. If the report is missing at the daily cutoff, the supervisor gets a reminder and the admin sees a flag. Submitting a report updates the resource data used by the feasibility agent and triggers a monitoring run for the affected orders.

5.8 Production tracking and the client progress view

The monitoring agent calculates, per order: current phase, percent complete per process and overall, produced versus total quantity, required daily output, actual rate, projected finish date, and a status of on track, at risk or behind.

- Admin: a tracking board with every order by phase, a detail page with the progress chart, daily reports and alerts. A behind-schedule alert includes recovery suggestions.
- Client: a timeline of phases (order received, quotation, confirmed, then each production stage, ready, delivered), the produced and remaining quantity, and an estimated completion date. Internal details are not shown. The page updates live.

5.9 Daily market price monitoring

Every day at the time set in Settings the market price agent:

1. Collects prices for the materials on the watchlist from the configured sources. Supplier quotations entered by the admin are the baseline.
2. Checks each observation against history and other sources. A large move from an unverified source is held for admin confirmation.
3. Calculates change versus yesterday, change over seven days and volatility, and compares them with the thresholds in Settings.
4. When a threshold is crossed, creates a price alert for the admin.

5. Finds the affected orders: sent or accepted quotations and confirmed orders whose materials are not yet purchased.
6. For orders with a material shortage, lists the best current purchase options by price and lead time. These feed feasibility recommendations and estimates.

The admin has a price centre with trend charts per material, the alert list and the affected orders.

5.10 Price surge and quotation revision

1. For each affected order the agent proposes a revision: old price, new price, percentage change, the cost drivers, any change to the delivery date, and a client-friendly reason such as 'cotton yarn rose 12% in seven days'.
2. The admin reviews, edits if needed, and approves or rejects. If rejected, the current price stands.
3. Only after approval is the client notified. The client sees the old and new price with the reason, and can accept, decline or ask questions.
4. If the client accepts, the order price is updated. If the client declines, the admin decides what happens next under the quotation terms.
5. Materials already bought are locked and excluded from revisions.

The quotation terms must include a price-variation clause so clients know a price can change when material costs move. Have this wording reviewed by a lawyer.

5.11 Chat and negotiation

Each order has a chat between the client and the AI assistant.

The assistant can	The assistant cannot
Explain why the price is what it is, using the approved reasons	Grant any discount or concession
Explain a price revision and the market movement behind it	Change a price, quantity or date
Share order status, phase and estimated completion	Share cost, margin or other clients' data
Explain the deadline and any risk factors in client-safe words	Promise anything that is not in the approved quotation

The assistant can	The assistant cannot
Answer general questions about the process	

When the client asks for a discount or anything extra:

1. The assistant tells the client it has passed the request to the admin.
2. An escalation is created with a summary: the request, the current price, and the conversation so far. The admin also sees internal context such as margin.
3. The admin is alerted and can reply in the same chat, approve a revised quotation through the revision flow, or decline the request.
4. The admin can take over the conversation at any time. The assistant pauses until it is handed back.

Every message is labelled by sender (client, assistant, admin). Figures in assistant replies are checked against the stored records, and anything the assistant is unsure about is escalated.

5.12 Delivery and payment

- Delivery: the admin records each delivery with date, quantity and invoice. Partial deliveries are supported. The client sees delivered quantities, and the order closes when everything is delivered and paid.
- Payment: the admin records invoices and receipts, and the system shows pending, partial or received. Automated payment reminders are not in the first release.

6. Notifications

Event	Admin	Supervisor	Client
New order submitted	Yes		
Feasibility report ready	Yes		
Quotation awaiting approval	Yes		
Quotation sent			Yes
Client accepted or declined	Yes		

Event	Admin	Supervisor	Client
Price alert, revision proposed	Yes		
Revised price sent			Yes
Discount request escalated	Yes		
Order assigned		Yes	Yes (status)
Daily report missing	Yes	Yes	
Behind schedule alert	Yes	Yes	
Order ready, delivered	Yes		Yes

Channels: in-app and email in the first release, with WhatsApp and SMS as provider-based additions.

7. Screens by role

Role	Screens
Client	Login, my orders, new order, order detail with progress timeline, quotation view with accept, decline and ask, chat, notifications
Admin	Dashboard and order pipeline, order inbox, feasibility report, quotation review and approval, price centre, purchase decisions and suppliers, chat inbox with escalations and takeover, orders tracking by phase, resources (materials, machines, workers), supervisor reports, clients, deliveries and payments, settings, users, audit, agent activity
Supervisor	Assigned orders, today's report form, report history, raise an issue, notifications

8. Business rules

ID	Rule
BR-1	A client sees a quotation only after the admin approves it.
BR-2	A price revision reaches the client only after the admin approves it.
BR-3	Discounts and extra demands are decided by the admin only.
BR-4	Every number comes from the domain engine, and figures in LLM text must match the records.
BR-5	A quotation includes wastage, a price-risk buffer, a validity date and the ETA basis.
BR-6	Feasibility accounts for the remaining load of confirmed orders.
BR-7	Materials already purchased are excluded from price revisions.
BR-8	Quotation terms include a price-variation clause.
BR-9	Only the admin assigns an order to supervisors.
BR-10	One daily report per supervisor per day. Corrections need a reason.
BR-11	Clients see client-safe information about their own orders only.
BR-12	Approvals, decisions and agent outputs are audited.
BR-13	Expired quotations cannot be accepted.
BR-14	A large price move from an unverified source needs admin confirmation before it can trigger revisions.

9. Exceptions and edge cases

Situation	Handling
Agent fails or times out	Record moves to a failed state with a retry button. Nothing hangs.
Little capacity data in the first weeks	Conservative buffers and a low-data warning on the report and quotation
Client changes the order after quoting	The admin creates a new version of the request. The old quotation is superseded and feasibility re-runs.
Admin edits an approved quotation after sending	A revision is created and sent again. The old one is superseded.
Supervisor misses the daily report	Reminder and flag. Monitoring uses the last known data and marks it stale.
Market source unavailable	Last known price is used and flagged stale
Assistant unsure of an answer	Escalates to the admin
Order cancelled	Stock reservations are released and supervisors notified

10. Open points

- The exact garment production stages and standard times, and whether machines are tracked individually or by type.
- Which market sources to use, and whether a paid price feed is acceptable.
- Whether the admin can proceed with an order marked at risk and accept the risk on record.
- Whether an advance payment will be recorded in a later release.